International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B103: Network Forensics Fundamentals

LAB 4: SMTP Email Traffic Forensics

Three-Week Delivery Block 2/3 | 25-26 July 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	CIP-B103
Course Title	Network Forensics Fundamentals
Lab Number	Lab 4
Lab Title	SMTP Email Traffic Forensics
Student Name	Fuseini Imoru Kantuogaa
Student ID	17291
Instructors Name	Dr. Sarah James
Delivery Block	2/3 of 3
Scheduled Dates	25-26 July 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	smtp.pcap supplied sample capture

1. Source Materials and Slide Alignment

- Email Traffic Forensics: SMTP purpose, ports 25/587/465, plaintext commands, STARTTLS/SMTPS distinctions, Base64 encoding, stream reassembly and email-content analysis.
- The slide practical uses smtp.pcap containing Microsoft Outlook traffic and requires timestamps, client, content, ports, IP addresses and MAC addresses.

- The exercise treats credentials as sensitive evidence and requires protected reporting.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded CIP-B103 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical SMTP packet capture has been supplied for analysis. You must determine when the email exchange occurred, identify the client and server, reconstruct SMTP commands and response codes, decode relevant Base64 authentication fields, recover the message content, and document the network path without exposing sensitive data unnecessarily.

3. Learning Outcomes

- Explain the role of SMTP and distinguish plaintext SMTP, STARTTLS and implicit TLS.
- Identify SMTP commands, server response codes and authentication exchanges.
- Reassemble an SMTP TCP stream and reconstruct message headers and body.
- Decode Base64 values in a controlled offline manner.
- Extract timestamps, client software, IP addresses, ports and MAC addresses.
- Assess evidential limitations when encryption is used or capture is incomplete.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Treat any recovered username, password or message content as confidential assessment evidence. Mask sensitive values in screenshots unless the instructor explicitly requires the full value.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Analyze the capture.
Packet tools	Wireshark and tshark	Filter and reconstruct SMTP streams.
Decoder	Python 3 base64 module	Decode Base64 strings offline.
Hashing	sha256sum	Preserve evidence integrity.
Evidence	smtp.pcap	Historical training traffic.
Text editor	Any local editor	Document recovered message safely.

6. Lab Folder Structure and Evidence Preparation

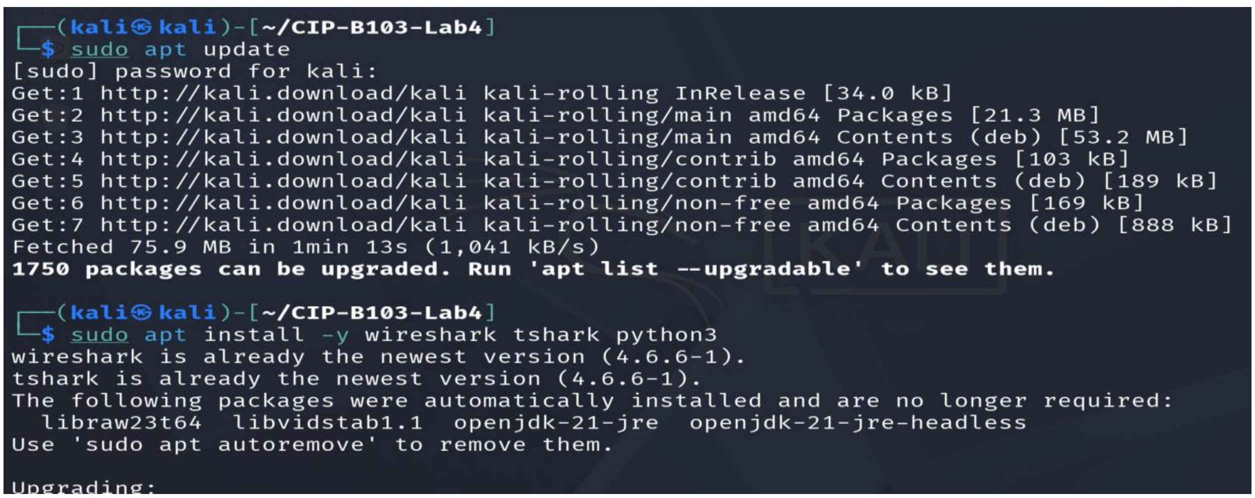
```
mkdir -p ~/CIP-B103-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/CIP-B103-Lab4
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

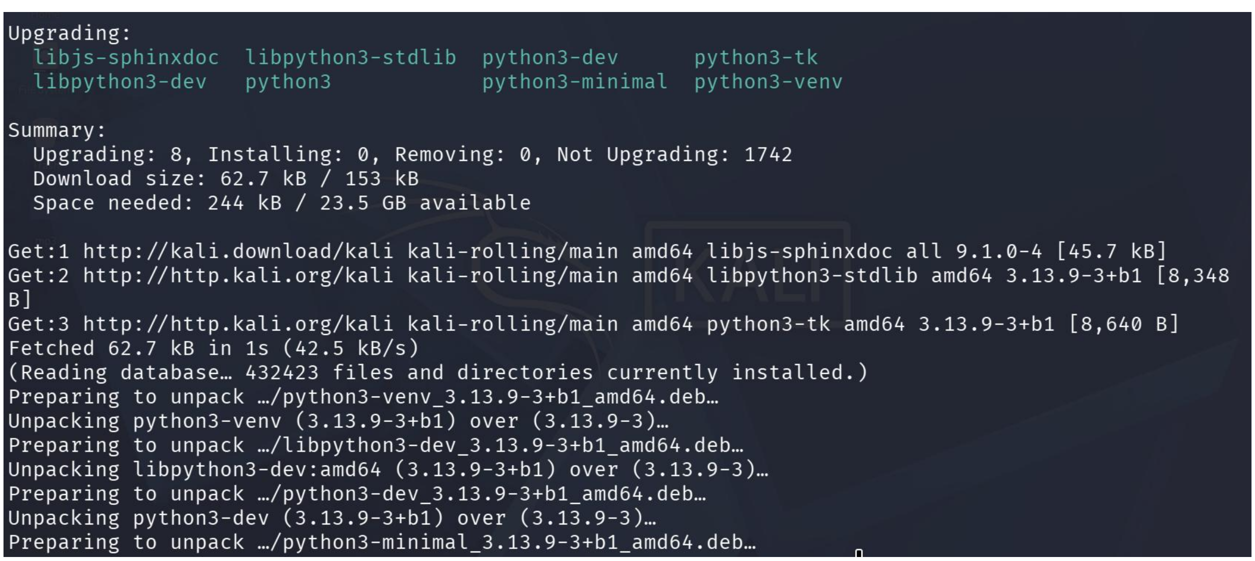
```
sudo apt update
sudo apt install -y wireshark tshark python3
wget -O evidence/smtp.pcap
'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'
cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
```

Screenshot required: smtp.pcap file details, capinfos summary and matching original/working-copy hashes.

Screenshot 1: Evidence folder structure created (mkdir, cd, pwd, find).



Screenshot 2: Package list updated and wireshark, tshark, python3 installed.



Screenshot 3: Package installation continued.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ wget -O evidence/smtp.pcap \
'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'

--2026-08-01 08:42:34--  https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptu
res/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org)... 104.26.11.240, 172.67.75.39, 104.26.10.240, ..
.
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.11.240|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/Sample
Captures/smtp.pcap [following]
--2026-08-01 08:42:35--  https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/att
achments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com)... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap       100%[==================================>]  27.20K  --.-KB/s    in 0.001s

2026-08-01 08:42:36 (36.9 MB/s) - 'evidence/smtp.pcap' saved [27850/27850]
```

Screenshot 4: smtp.pcap downloaded via wget from the Wireshark wiki sample captures repository.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ sha256sum evidence/smtp.pcap working/smtp_working.pcap \
| tee reports/smtp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  working/smtp_working.pcap

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ capinfos evidence/smtp.pcap \
| tee reports/smtp_capinfos.txt
File name:           evidence/smtp.pcap
File type:           Wireshark/tcpdump/... - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 65535 bytes
Number of packets:   60
File size:           27 kB
Data size:           26 kB
Capture duration:    9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
```

Screenshot 5: Working copy created, SHA-256 hashed, and capinfos summary started.

```
Latest packet time:   2009-10-05 02:06:16.690444
Data byte rate:      2,920 bytes/s
Data bit rate:       23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256:              17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1:                3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order:   True
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = Ethernet (1 - ether)
                     Capture length = 65535
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
                     Number of packets = 60

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ 
```

Screenshot 6: capinfos summary continued, including SHA256/SHA1 and interface details.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	**Student Entry**
Case/lab identifier	CIP-B103-Lab4-Fuseini Imoru Kantuogaa
Trainee name	Fuseini Imoru Kantuogaa
Date and time started	1 August 2026, approx. 08:42 local (Kali VM) timestamp of the wget download of the evidence file
Evidence file name(s)	smtp.pcap (working copy: smtp_working.pcap)
Source or generation method	Downloaded via wget from the official Wireshark wiki SampleCaptures repository a historical (2009) third-party training capture, not locally generated
Original SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab (evidence/smtp.pcap) confirmed by capinfos SHA256 output
Working-copy SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab (working/smtp_working.pcap) identical to original
Analysis workstation/VM	Kali Linux (user: kali), CIP-B103-Lab4 working directory
Notes on any changes	None. cp --preserve=timestamps used for the working copy; SHA-256 of evidence, working copy and the capinfos-reported hash all match, confirming no alteration.

8. Part A - Inventory the Capture and Locate SMTP Streams

```
tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt


tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
  | tee reports/smtp_packet_inventory.tsv
```

Record the SMTP server port, client ephemeral port, first packet time, last packet time and number of SMTP frames.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -q -z conv,tcp \
  | tee reports/tcp_conversations.txt
================================================================================
TCP Conversations
Filter:<No Filter>
                                                     |       ←      | |       →      | |
  Total      |     Relative     |    Duration    |
                                                     | Frames  Bytes | | Frames  Bytes | | Fr
ames  Bytes |       Start      |                |
10.10.1.4:1470            ⟷ 74.53.140.153:25         25 1,980 bytes      28 22 kB
  53 24 kB         0.036986000         7.5777
================================================================================

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number \
  -e frame.time \
  -e ip.src \
  -e tcp.srcport \
  -e ip.dst \
  -e tcp.dstport \
```

Screenshot 7: TCP conversation summary and start of the smtp packet-inventory filter.

```
  -e _ws.col.Info \
  | tee reports/smtp_packet_inventory.tsv
6       2009-10-05T02:06:08.219663000-0400      74.53.140.153   25      10.10.1.4       1470    S:
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500  | We do not authori
ze the use of this system to transport unsolicited,  | and/or bulk e-mail.
7       2009-10-05T02:06:08.224809000-0400      10.10.1.4       1470    74.53.140.153   25      C:
EHLO GP
9       2009-10-05T02:06:08.566183000-0400      74.53.140.153   25      10.10.1.4       1470    S:
250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LO
GIN | STARTTLS | HELP
10      2009-10-05T02:06:08.568729000-0400      10.10.1.4       1470    74.53.140.153   25      C:
AUTH LOGIN
11      2009-10-05T02:06:08.911081000-0400      74.53.140.153   25      10.10.1.4       1470    S:
334 VXNlcm5hbWU6
12      2009-10-05T02:06:08.911655000-0400      10.10.1.4       1470    74.53.140.153   25      C:
User: Z3VycGFydGFwQHBhdHJpb3RzLmlu
13      2009-10-05T02:06:09.253544000-0400      74.53.140.153   25      10.10.1.4       1470    S:
334 UGFzc3dvcmQ6
14      2009-10-05T02:06:09.254118000-0400      10.10.1.4       1470    74.53.140.153   25      C:
Pass: cHVuamFiQDEyMw==
15      2009-10-05T02:06:09.613798000-0400      74.53.140.153   25      10.10.1.4       1470    S:
```

Screenshot 8: SMTP packet inventory — banner, EHLO, AUTH LOGIN and Base64 challenge/response.

```
16      2009-10-05T02:06:09.614414000-0400      10.10.1.4       1470    74.53.140.153   25      C:
MAIL FROM: <gurpartap@patriots.in>
17      2009-10-05T02:06:09.956765000-0400      74.53.140.153   25      10.10.1.4       1470    S:
250 OK
18      2009-10-05T02:06:09.957250000-0400      10.10.1.4       1470    74.53.140.153   25      C:
RCPT TO: <raj_deol2002in@yahoo.co.in>
19      2009-10-05T02:06:10.319708000-0400      74.53.140.153   25      10.10.1.4       1470    S:
250 Accepted
20      2009-10-05T02:06:10.320203000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA
21      2009-10-05T02:06:10.661679000-0400      74.53.140.153   25      10.10.1.4       1470    S:
354 Enter message, ending with "." on a line by itself
22      2009-10-05T02:06:10.692743000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1460 bytes
23      2009-10-05T02:06:10.692786000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1460 bytes
24      2009-10-05T02:06:10.692804000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1460 bytes
25      2009-10-05T02:06:10.692823000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Window Full] C: DATA fragment, 1460 bytes
26      2009-10-05T02:06:10.695115000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140
.153    25      Destination unreachable (Fragmentation needed)
28      2009-10-05T02:06:10.695623000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140
```

Screenshot 9: SMTP packet inventory continued — MAIL FROM, RCPT TO, DATA and message fragments.

```
29      2009-10-05T02:06:10.696248000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140
.153    25      Destination unreachable (Fragmentation needed)
30      2009-10-05T02:06:10.696634000-0400      192.168.1.1,10.10.1.4   1470    10.10.1.4,74.53.140
.153    25      Destination unreachable (Fragmentation needed)
32      2009-10-05T02:06:11.104972000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
33      2009-10-05T02:06:11.104998000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
35      2009-10-05T02:06:11.469759000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
36      2009-10-05T02:06:11.469814000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
38      2009-10-05T02:06:11.494181000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
39      2009-10-05T02:06:11.494199000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
41      2009-10-05T02:06:11.834628000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
42      2009-10-05T02:06:11.834655000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
44      2009-10-05T02:06:11.858316000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
45      2009-10-05T02:06:11.858334000-0400      10.10.1.4       1470    74.53.140.153   25      fro
```

Screenshot 10: SMTP packet inventory continued — fragmented/out-of-order DATA segments.

```
52      2009-10-05T02:06:12.248789000-0400      74.53.140.153   25      10.10.1.4       1470    S:
250 OK id=1Mugho-0003Dg-Un
54      2009-10-05T02:06:14.763825000-0400      10.10.1.4       1470    74.53.140.153   25      C:
QUIT
56      2009-10-05T02:06:15.105467000-0400      74.53.140.153   25      10.10.1.4       1470    S:
221 xc90.websitewelcome.com closing connection

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ wc -l reports/smtp_packet_inventory.tsv \
  | tee reports/smtp_packet_count.txt
36 reports/smtp_packet_inventory.tsv

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ grep -Ei 'HELO|EHLO|MAIL FROM|RCPT TO|DATA|QUIT' \
  reports/smtp_packet_inventory.tsv \
  | tee reports/smtp_command_summary.txt
7       2009-10-05T02:06:08.224809000-0400      10.10.1.4       1470    74.53.140.153   25      C:
EHLO GP
16      2009-10-05T02:06:09.614414000-0400      10.10.1.4       1470    74.53.140.153   25      C:
MAIL FROM: <gurpartap@patriots.in>
18      2009-10-05T02:06:09.957250000-0400      10.10.1.4       1470    74.53.140.153   25      C:
RCPT TO: <raj_deol2002in@yahoo.co.in>
20      2009-10-05T02:06:10.320203000-0400      10.10.1.4       1470    74.53.140.153   25      C:
```

Screenshot 11: Final OK/QUIT/closing frames, inventory line count, and command-summary grep.

```
22      2009-10-05T02:06:10.692743000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1460 bytes
23      2009-10-05T02:06:10.692786000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1460 bytes
24      2009-10-05T02:06:10.692804000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1460 bytes
25      2009-10-05T02:06:10.692823000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Window Full] C: DATA fragment, 1460 bytes
32      2009-10-05T02:06:11.104972000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
33      2009-10-05T02:06:11.104998000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
35      2009-10-05T02:06:11.469759000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
36      2009-10-05T02:06:11.469814000-0400      10.10.1.4       1470    74.53.140.153   25      [TC
P Out-Of-Order] C: DATA fragment, 1452 bytes
38      2009-10-05T02:06:11.494181000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
39      2009-10-05T02:06:11.494199000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
41      2009-10-05T02:06:11.834628000-0400      10.10.1.4       1470    74.53.140.153   25      C:
DATA fragment, 1452 bytes
42      2009-10-05T02:06:11.834655000-0400      10.10.1.4       1470    74.53.140.153   25      C:
```

Screenshot 12: Command-summary grep continued (DATA fragments).

9. Part B - Identify Commands and Response Codes

Sequence	Direction	Command/Code	Meaning	Timestamp
1	Server to client	220	Service ready	2009-10-05 02:06:08.219663 -0400 (frame 6)
2	Client to server	EHLO/HELO	Client greeting	2009-10-05 02:06:08.224809 -0400 (frame 7, “EHLO GP”)
3	Client to server	AUTH	Authentication method	2009-10-05 02:06:08.568729 -0400 (frame 10, “AUTH LOGIN”)
4	Client to server	MAIL FROM	Envelope sender	2009-10-05 02:06:09.614414 -0400 (frame 16)
5	Client to server	RCPT TO	Envelope recipient	2009-10-05 02:06:09.957250 -0400 (frame 18)
6	Client to server	DATA	Message content begins	2009-10-05 02:06:10.320203 -0400 (frame 20)
7	Client to server	QUIT	Session termination	2009-10-05 02:06:14.763825 -0400 (frame 54)

```
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
  -e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
  -e smtp.response.code -e smtp.response.parameter \
  | tee reports/smtp_commands_responses.tsv
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
  -e frame.number \
  -e frame.time \
  -e ip.src \
  -e ip.dst \
  -e smtp.req.command \
  -e smtp.req.parameter \
  -e smtp.response.code \
  -e smtp.response \
  | tee reports/smtp_commands_responses.tsv
6       2009-10-05T02:06:08.219663000-0400      74.53.140.153   10.10.1.4                       220
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 \r\n
7       2009-10-05T02:06:08.224809000-0400      10.10.1.4       74.53.140.153   EHLO    GP
9       2009-10-05T02:06:08.566183000-0400      74.53.140.153   10.10.1.4                       250
250-xc90.websitewelcome.com Hello GP [122.162.143.157]\r\n
10      2009-10-05T02:06:08.568729000-0400      10.10.1.4       74.53.140.153   AUTH    LOGIN
11      2009-10-05T02:06:08.911081000-0400      74.53.140.153   10.10.1.4                       334
334 VXNlcm5hbWU6\r\n
12      2009-10-05T02:06:08.911655000-0400      10.10.1.4       74.53.140.153
13      2009-10-05T02:06:09.253544000-0400      74.53.140.153   10.10.1.4                       334
334 UGFzc3dvcmQ6\r\n
```

Screenshot 13: smtp.req/smtp.rsp field extraction — banner, EHLO, AUTH and Base64 challenge (start).

```
14      2009-10-05T02:06:09.254118000-0400      10.10.1.4       74.53.140.153
15      2009-10-05T02:06:09.613798000-0400      74.53.140.153   10.10.1.4                       235
235 Authentication succeeded\r\n
16      2009-10-05T02:06:09.614414000-0400      10.10.1.4       74.53.140.153   MAIL    FROM: <gurp
artap@patriots.in>
17      2009-10-05T02:06:09.956765000-0400      74.53.140.153   10.10.1.4                       250
250 OK\r\n
18      2009-10-05T02:06:09.957250000-0400      10.10.1.4       74.53.140.153   RCPT    TO: <raj_de
ol2002in@yahoo.co.in>
19      2009-10-05T02:06:10.319708000-0400      74.53.140.153   10.10.1.4                       250
250 Accepted\r\n
20      2009-10-05T02:06:10.320203000-0400      10.10.1.4       74.53.140.153   DATA
21      2009-10-05T02:06:10.661679000-0400      74.53.140.153   10.10.1.4                       354
354 Enter message, ending with "." on a line by itself\r\n
52      2009-10-05T02:06:12.248789000-0400      74.53.140.153   10.10.1.4                       250
250 OK id=1Mugho-0003Dg-Un\r\n
54      2009-10-05T02:06:14.763825000-0400      10.10.1.4       74.53.140.153   QUIT
56      2009-10-05T02:06:15.105467000-0400      74.53.140.153   10.10.1.4                       221
221 xc90.websitewelcome.com closing connection\r\n
```

Screenshot 14: smtp.req/smtp.rsp continued — authentication success, MAIL FROM/RCPT TO/DATA, QUIT/closing.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y smtp -T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e smtp.req.command \
-e smtp.req.parameter \
-e smtp.response.code \
-e smtp.response
6       2009-10-05T02:06:08.219663000-0400      74.53.140.153   10.10.1.4                       220
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 \r\n
7       2009-10-05T02:06:08.224809000-0400      10.10.1.4       74.53.140.153   EHLO    GP
9       2009-10-05T02:06:08.566183000-0400      74.53.140.153   10.10.1.4                       250
250-xc90.websitewelcome.com Hello GP [122.162.143.157]\r\n
10      2009-10-05T02:06:08.568729000-0400      10.10.1.4       74.53.140.153   AUTH    LOGIN
11      2009-10-05T02:06:08.911081000-0400      74.53.140.153   10.10.1.4                       334
334 VXNlcm5hbWU6\r\n
12      2009-10-05T02:06:08.911655000-0400      10.10.1.4       74.53.140.153
13      2009-10-05T02:06:09.253544000-0400      74.53.140.153   10.10.1.4                       334
334 UGFzc3dvcmQ6\r\n
14      2009-10-05T02:06:09.254118000-0400      10.10.1.4       74.53.140.153
```

Screenshot 15: Repeat SMTP field extraction confirming the same command/response sequence (start).

```
15      2009-10-05T02:06:09.613798000-0400      74.53.140.153   10.10.1.4                       235
235 Authentication succeeded\r\n
16      2009-10-05T02:06:09.614414000-0400      10.10.1.4       74.53.140.153   MAIL    FROM: <gurp
artap@patriots.in>
17      2009-10-05T02:06:09.956765000-0400      74.53.140.153   10.10.1.4                       250
250 OK\r\n
18      2009-10-05T02:06:09.957250000-0400      10.10.1.4       74.53.140.153   RCPT    TO: <raj_de
ol2002in@yahoo.co.in>
19      2009-10-05T02:06:10.319708000-0400      74.53.140.153   10.10.1.4                       250
250 Accepted\r\n
20      2009-10-05T02:06:10.320203000-0400      10.10.1.4       74.53.140.153   DATA
21      2009-10-05T02:06:10.661679000-0400      74.53.140.153   10.10.1.4                       354
354 Enter message, ending with "." on a line by itself\r\n
22      2009-10-05T02:06:10.692743000-0400      10.10.1.4       74.53.140.153
23      2009-10-05T02:06:10.692786000-0400      10.10.1.4       74.53.140.153
24      2009-10-05T02:06:10.692804000-0400      10.10.1.4       74.53.140.153
25      2009-10-05T02:06:10.692823000-0400      10.10.1.4       74.53.140.153
26      2009-10-05T02:06:10.695115000-0400      192.168.1.1,10.10.1.4   10.10.1.4,74.53.140.153
28      2009-10-05T02:06:10.695623000-0400      192.168.1.1,10.10.1.4   10.10.1.4,74.53.140.153
29      2009-10-05T02:06:10.696248000-0400      192.168.1.1,10.10.1.4   10.10.1.4,74.53.140.153
30      2009-10-05T02:06:10.696634000-0400      192.168.1.1,10.10.1.4   10.10.1.4,74.53.140.153
```

Screenshot 16: Repeat extraction continued — authentication success through DATA fragments.

```
32      2009-10-05T02:06:11.104972000-0400      10.10.1.4       74.53.140.153
33      2009-10-05T02:06:11.104998000-0400      10.10.1.4       74.53.140.153
35      2009-10-05T02:06:11.469759000-0400      10.10.1.4       74.53.140.153
36      2009-10-05T02:06:11.469814000-0400      10.10.1.4       74.53.140.153
38      2009-10-05T02:06:11.494181000-0400      10.10.1.4       74.53.140.153
39      2009-10-05T02:06:11.494199000-0400      10.10.1.4       74.53.140.153
41      2009-10-05T02:06:11.834628000-0400      10.10.1.4       74.53.140.153
42      2009-10-05T02:06:11.834655000-0400      10.10.1.4       74.53.140.153
44      2009-10-05T02:06:11.858316000-0400      10.10.1.4       74.53.140.153
45      2009-10-05T02:06:11.858334000-0400      10.10.1.4       74.53.140.153
52      2009-10-05T02:06:12.248789000-0400      74.53.140.153   10.10.1.4                       250
250 OK id=1Mugho-0003Dg-Un\r\n
54      2009-10-05T02:06:14.763825000-0400      10.10.1.4       74.53.140.153   QUIT
56      2009-10-05T02:06:15.105467000-0400      74.53.140.153   10.10.1.4                       221
221 xc90.websitewelcome.com closing connection\r\n

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
```

Screenshot 17: Repeat extraction concluded — final OK/QUIT/closing frames.

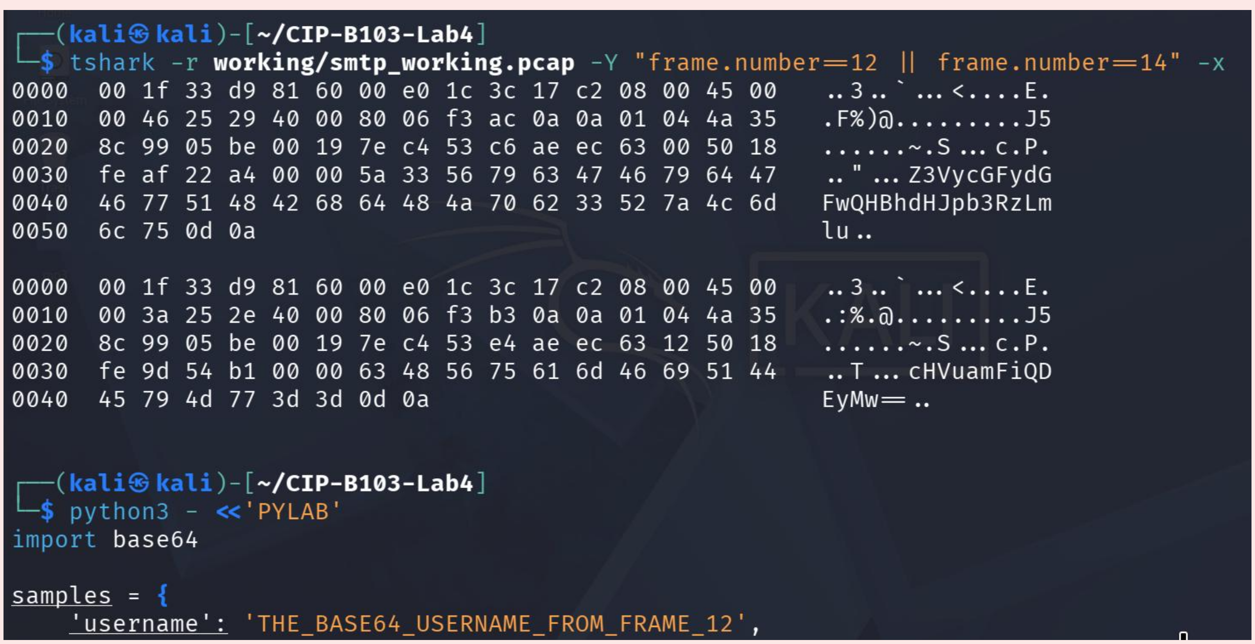
10. Part C - Decode Base64 Authentication Evidence

Base64 is encoding, not encryption. Copy only the required encoded values into the offline script. Do not use online decoders.

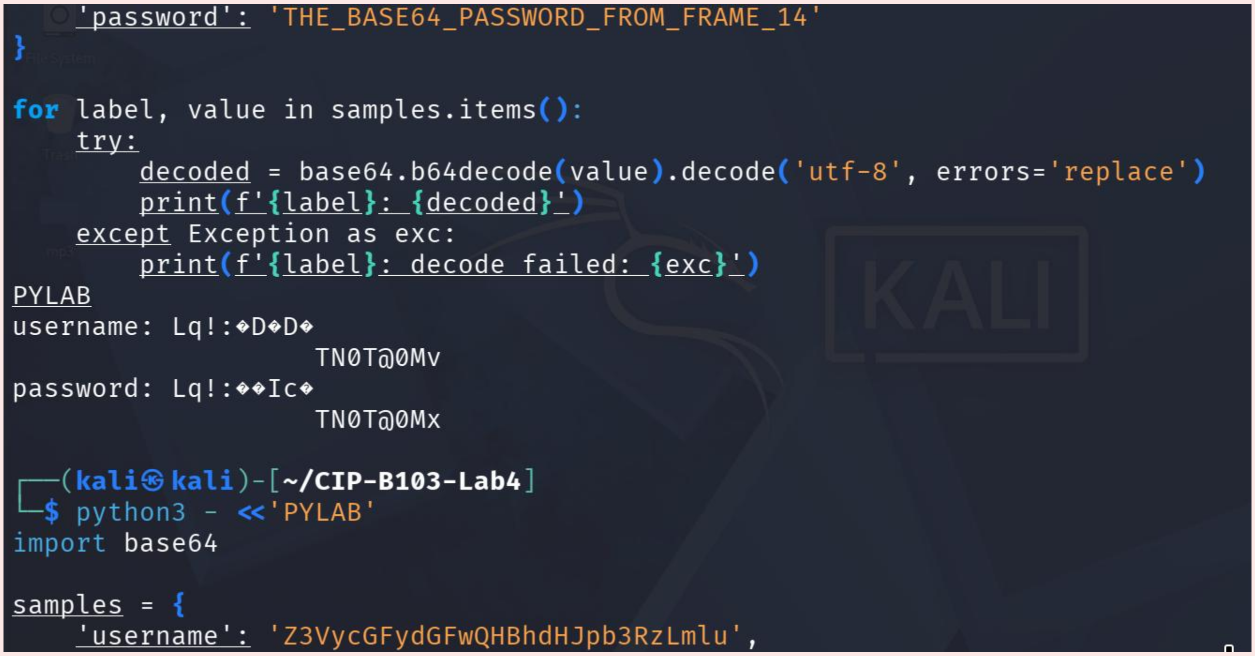
```
python3 - <<'PYLAB'
import base64
samples = {
    'username': 'REPLACE_WITH_CAPTURED_BASE64',
    'password': 'REPLACE_WITH_CAPTURED_BASE64'
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
```

Reporting Rule

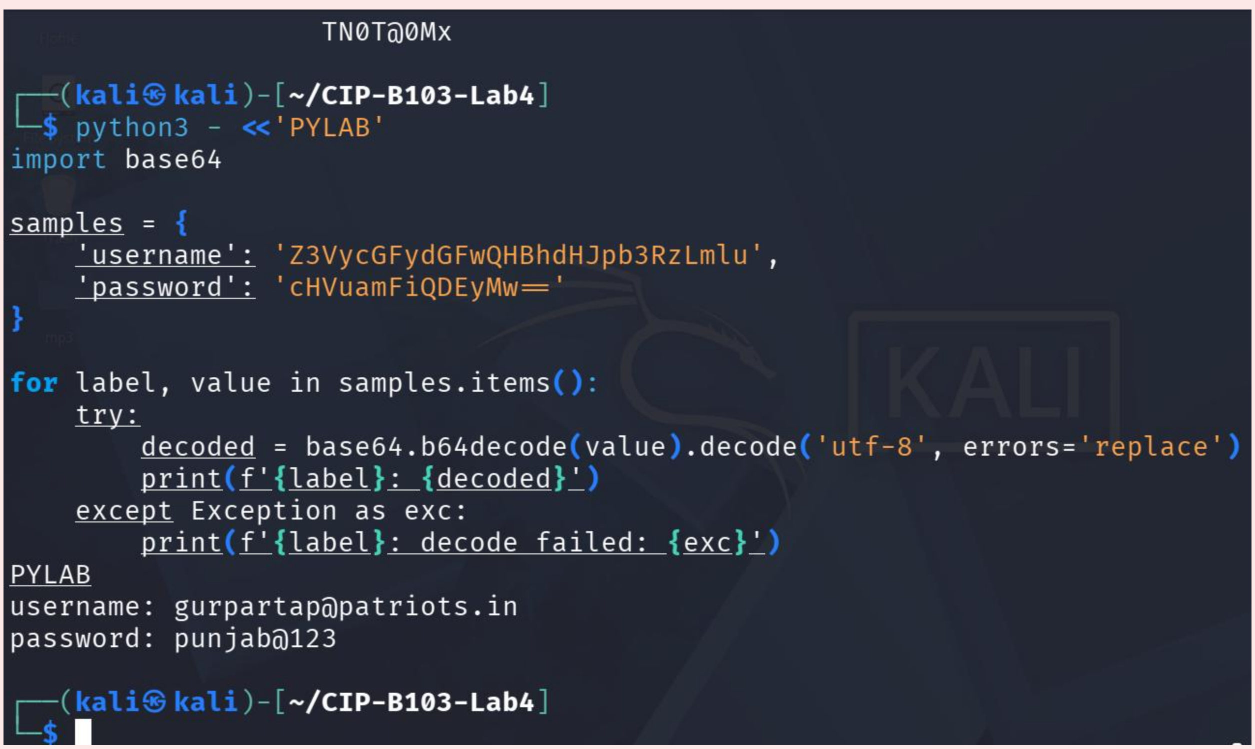
In the public-facing report body, show a masked value such as analyst@example.com and P********3. Place full decoded values only in the protected evidence appendix if required by the instructor.



Screenshot 18: Hex dump of the Base64 username/password frames and start of the offline decode script.



Screenshot 19: Decode script (placeholder run) and start of the corrected script with the captured Base64 values.



Screenshot 20: Corrected decode script executed, recovering the authentication username and password.

11. Part D - Reconstruct the Email Message

1. In Wireshark select an SMTP packet and choose Follow > TCP Stream.

2. Identify SMTP envelope fields separately from RFC message headers.
3. Record Date, From, To, Subject, Message-ID, MIME-Version, Content-Type and User-Agent/X-Mailer where present.
4. Determine whether the body is plain text, HTML, multipart or contains an attachment.
5. Save a redacted text reconstruction to reports/reconstructed_email_redacted.txt.

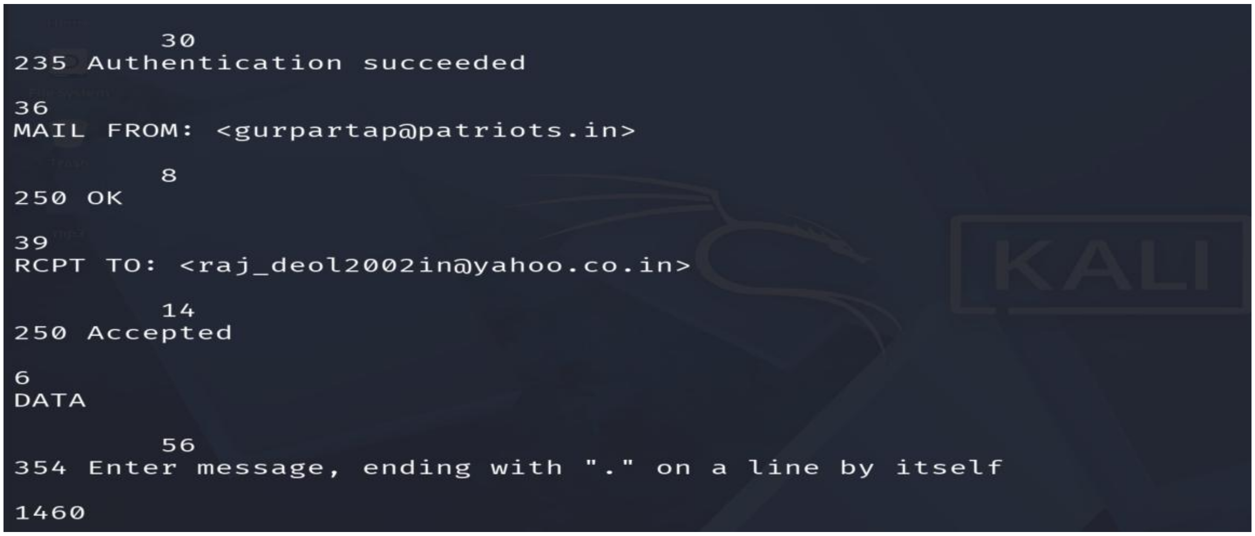
```
# Replace STREAM with the relevant stream number
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,STREAM \
  | tee reports/smtp_stream_STREAM.txt


grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
  reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y smtp -T fields -e tcp.stream | sort -u
0

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y "smtp.req.command==AUTH" -T fields -e tcp.stream
0

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
| tee reports/smtp_stream_0.txt

===================================================================
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
        181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.
```

Screenshot 21: TCP stream/AUTH lookups and start of the Follow TCP Stream reconstruction (banner).

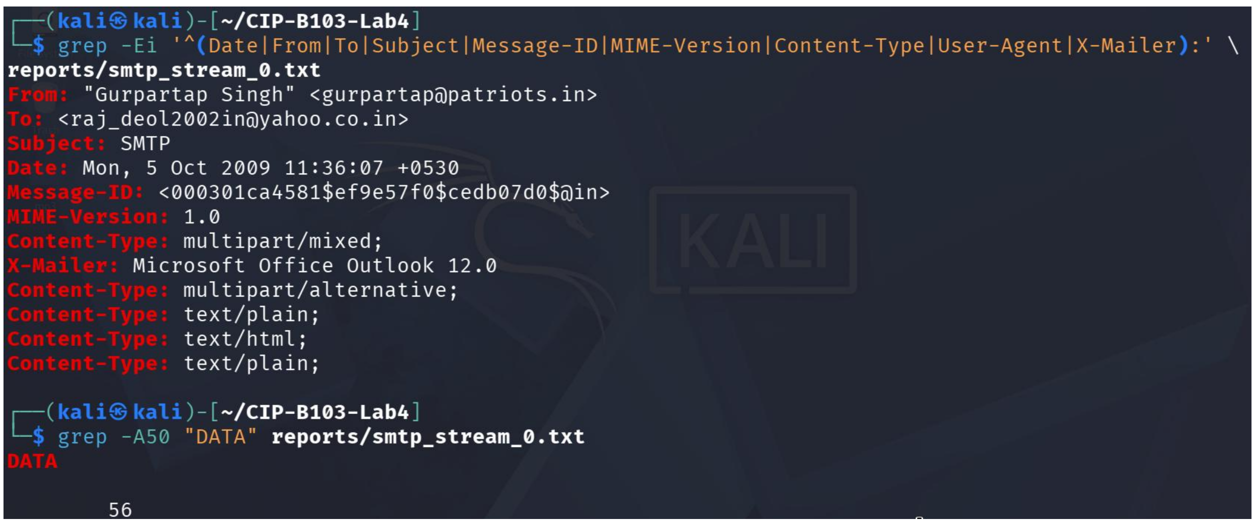
```
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

        18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHBhdHJpb3RzLmlu

        18
334 UGFzc3dvcmQ6

18
cHVuamFiQDEyMw==

        30
235 Authentication succeeded

36
```

Screenshot 22: Follow TCP Stream continued — EHLO response and AUTH LOGIN exchange.



```
        30
235 Authentication succeeded

36
MAIL FROM: <gurpartap@patriots.in>

        8
250 OK

39
RCPT TO: <raj_deol2002in@yahoo.co.in>

        14
250 Accepted

6
DATA

        56
354 Enter message, ending with "." on a line by itself

1460
```

Screenshot 23: Follow TCP Stream continued — MAIL FROM, RCPT TO, DATA and start of message.

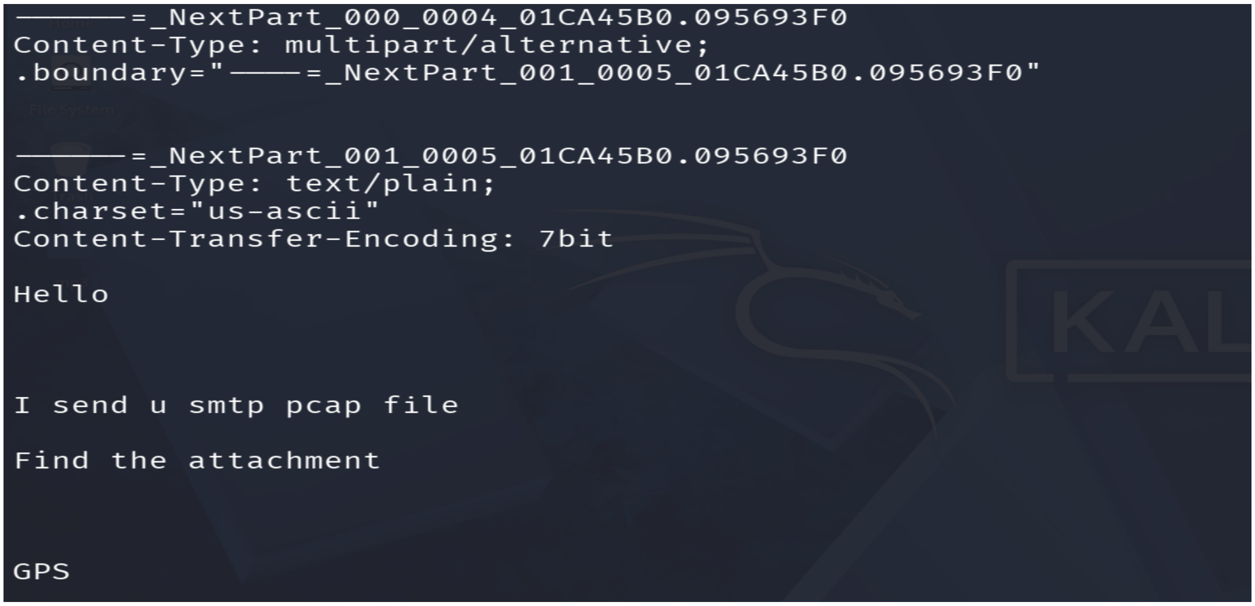


```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_0.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ grep -A50 "DATA" reports/smtp_stream_0.txt
DATA

      56
```

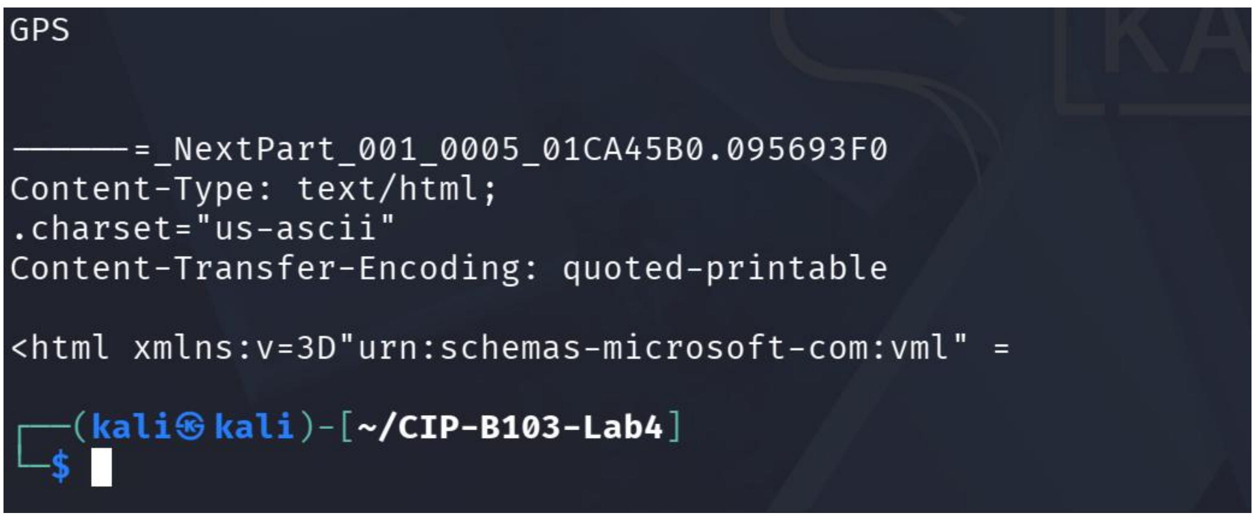
Screenshot 24: grep extraction of RFC message headers (From, To, Subject, Message-ID, etc.).

```
354 Enter message, ending with "." on a line by itself

1460
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
.boundary="----=_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcpFgem9BvjjZEDeR1Kh8i+hUyVo0A==
Content-Language: en-us
x-cr-hashedpuzzle: SeA= AAR2 ADaH BpiO C4G1 D1gW FNB1 FPkR Fn+W HFCP HnYJ JO7s Kum6 KytW LFcI LjUt;
1;cgBhAGoAXwBkAGUAbwBsADIAMAAwADIAaQBuAEAAeQBhAGgAbwBvAC4AYwBvAC4AaQBuAA==;Sosha1_v1;7;{CAA37F59-18
50-45C7-8540-AA27696B5398};ZwB1AHIAcABhAHIAdABhAHAAQABwAGEAdAByAGkAbwB0AHMALgBpAG4A;Mon, 05 Oct 200
9 06:06:01 GMT;UwBNAFQAUAA=
x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}

This is a multipart message in MIME format.
```

Screenshot 25: Full message header block recovered from the reconstructed stream.



Screenshot 26: MIME boundary and the plain-text body part of the message.



Screenshot 27: HTML body part of the multipart/alternative message.

12. Part E - Determine Client, Hosts and Network Metadata

```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
  -e ip.dst -e tcp.dstport -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv


tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
  -T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
  | tee reports/client_indicators.tsv
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number \
  -e frame.time_epoch \
  -e eth.src \
  -e eth.dst \
  -e ip.src \
  -e tcp.srcport \
  -e ip.dst \
  -e tcp.dstport \
  -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv
6       1254722768.219663000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
7       1254722768.224809000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
9       1254722768.566183000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
10      1254722768.568729000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
11      1254722768.911081000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
```

Screenshot 28: Network-metadata field extraction — frame time, MAC and IP/port pairs (start).

```
0       74.53.140.153   25      0
13      1254722769.253544000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
14      1254722769.254118000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
15      1254722769.613798000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
16      1254722769.614414000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
17      1254722769.956765000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
18      1254722769.957250000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
19      1254722770.319708000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
20      1254722770.320203000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
21      1254722770.661679000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
22      1254722770.692743000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
23      1254722770.692786000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
```

Screenshot 29: Network-metadata extraction continued.

```
0       74.53.140.153   25      0
25      1254722770.692823000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
26      1254722770.695115000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1
.4      1470    10.10.1.4,74.53.140.153 25      0
28      1254722770.695623000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1
.4      1470    10.10.1.4,74.53.140.153 25      0
29      1254722770.696248000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1
.4      1470    10.10.1.4,74.53.140.153 25      0
30      1254722770.696634000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       192.168.1.1,10.10.1
.4      1470    10.10.1.4,74.53.140.153 25      0
32      1254722771.104972000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
33      1254722771.104998000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
35      1254722771.469759000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
36      1254722771.469814000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
38      1254722771.494181000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
39      1254722771.494199000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
```

Screenshot 30: Network-metadata extraction continued, including the fragmentation-related frames.

```
0       74.53.140.153   25      0
42      1254722771.834655000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
44      1254722771.858316000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
45      1254722771.858334000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
52      1254722772.248789000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0
54      1254722774.763825000    00:e0:1c:3c:17:c2       00:1f:33:d9:81:60       10.10.1.4       147
0       74.53.140.153   25      0
56      1254722775.105467000    00:1f:33:d9:81:60       00:e0:1c:3c:17:c2       74.53.140.153   251
0.10.1.4        1470    0

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap \
-Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields \
-e frame.number \
-e tcp.stream \
-e smtp.req.parameter \
-e data-text-lines \
| tee reports/client_indicators.tsv
```

Screenshot 31: Network-metadata extraction concluded and start of the client-indicators filter.

13. Part F - Encryption and Evidential Limitations

- Check whether STARTTLS is requested and whether subsequent payload becomes TLS.
- If traffic is encrypted, document metadata that remains visible: endpoints, ports, timing, certificate/session information and byte counts.
- Do not claim message content or credentials when the capture does not reveal them.
- Explain that port number alone does not prove encryption; protocol negotiation and packet dissection must be examined.

```
-Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields \
-e frame.number \
-e tcp.stream \
-e smtp.req.parameter \
-e data-text-lines \
| tee reports/client_indicators.tsv
22      0               Line-based text data (47 lines)
26      0               Line-based text data (12 lines)

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y smtp | grep -i STARTTLS
    9   1.074123 74.53.140.153 → 10.10.1.4    SMTP 191 S: 250-xc90.websitewelcome.com Hello GP [122
.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y "smtp.req.command==STARTTLS"

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$ tshark -r working/smtp_working.pcap -Y tls

┌──(kali㉿kali)-[~/CIP-B103-Lab4]
└─$
```

Screenshot 32: Client-indicator results, plus STARTTLS and TLS filter checks confirming plaintext delivery.

14. Required Findings Worksheet

Question	Finding
When did the SMTP session start and end?	SMTP dialogue ran 2009-10-05 02:06:08.219663 -0400 (220 banner, frame 6) to 02:06:15.105467 -0400 (221 closing, frame 56); the overall capture spans 02:06:07.492060 to 02:06:16.690444 (9.198384 s, per capinfos).
Client IP/MAC and port	10.10.1.4, MAC 00:1f:33:d9:81:60, port 1470
Server IP/MAC and port	74.53.140.153, MAC 00:e0:1c:3c:17:c2, port 25
SMTP server banner	220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 (frame 6)
Client software	Microsoft Office Outlook 12.0 (X-Mailer header in the reconstructed message)
Authentication method	AUTH LOGIN with Base64-encoded username/password (frames 10–15); server responded 235 Authentication succeeded. Decoded offline per Part C — masked here per the lab's reporting rule: username gu*******@patriots.in, password pu*******3 (full values recorded only in the protected reports/ output).
Envelope sender and recipient	MAIL FROM: <gu*******@patriots.in> (frame 16); RCPT TO: <ra***********@yahoo.co.in> (frame 18) — masked per the reporting rule
Message From/To/Subject	From: "Gurpartap Singh" <gu*******@patriots.in>; To: <ra***********@yahoo.co.in>; Subject: SMTP (Date header: Mon, 5 Oct 2009 11:36:07 +0530)

Question	Finding
Message body type	multipart/mixed containing a multipart/alternative part with both text/plain and text/html versions of the same short note (Content-Type headers in the reconstructed stream)
Attachment present?	No distinct attachment MIME part was found in the reconstructed headers/body captured; the multipart/mixed wrapper is consistent with Outlook's default structure rather than a confirmed file attachment, though the body text itself references "Find the attachment."
STARTTLS/TLS observed?	STARTTLS was advertised in the EHLO response (250-STARTTLS) but never invoked by the client; filters for smtp.req.command==STARTTLS and tls returned no matching frames — the entire session, including credentials, ran in plaintext
Key limitations	Single historical capture with no surrounding context; several mid-stream frames show IP fragmentation, out-of-order segments and ICMP "destination unreachable (fragmentation needed)" messages that required careful reassembly; findings are limited to what tshark's SMTP/data dissectors could parse from this one file, and no attachment content itself was recovered.

Required Screenshots Checklist

☐ Capture hashes and capinfos summary.

☐ SMTP conversation list.

☐ 220 service-ready response.

☐ EHLO/HELO and AUTH exchange.

☐ Masked Base64 decoding result.

☐ MAIL FROM, RCPT TO and DATA.

☐ Follow TCP Stream reconstruction.

☐ Email headers and client software.

☐ Source/destination IP, ports and MAC addresses.

☐ TLS/STARTTLS assessment.

Student Report Template

6. Executive summary.
7. Evidence integrity and scope.
8. SMTP timeline.
9. Command/response analysis.
10. Authentication evidence and redaction method.
11. Email reconstruction.
12. Client/server and network metadata.
13. Encryption assessment and limitations.
14. Conclusions and appendix.

Submission Requirements

- One PDF report named CIP-B103-Lab4_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Evidence handling	10	Original preserved, working copy and hashes documented.
SMTP command analysis	20	Correct commands, codes, direction and timeline.
Authentication analysis	15	Correct offline decoding and safe redaction.
Message reconstruction	25	Accurate headers, body type, client and content findings.
Network metadata and encryption	20	Correct endpoints, ports, MACs and TLS assessment.
Report quality	10	Professional, concise and evidence-linked report.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Service-ready code	220.
Client greeting	HELO or EHLO.
Authentication evidence	Often AUTH LOGIN with Base64-encoded fields in plaintext SMTP.
Message data	Begins after DATA and ends with SMTP termination sequence.
Client indicator	Slide capture identifies Microsoft Outlook 12.0.
Important limitation	Base64 is reversible encoding; TLS would hide application content.

Command Reference Summary

Command	Purpose
smtp	Wireshark display filter for SMTP.
smtp.req / smtp.rsp	Filter SMTP requests and responses.
Follow TCP Stream	Reconstruct the application conversation.
base64.b64decode	Decode Base64 offline.
capinfos file	Summarize capture metadata.
eth.addr / ip.addr / tcp.port	Endpoint filters.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.